Estimating in Building Construction

Chapter 4 The Estimate

Organization

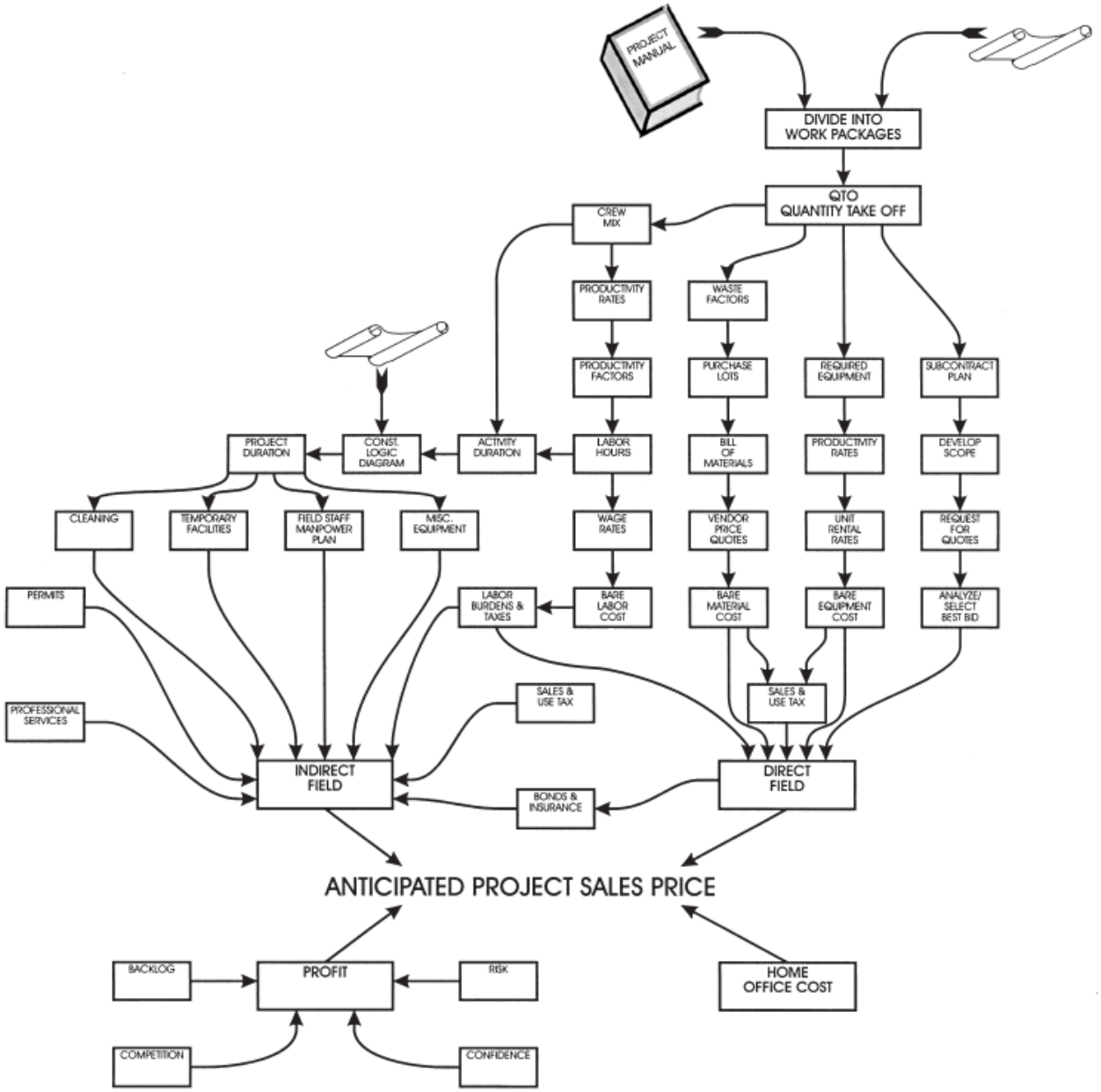
- The estimator must maintain a high degree of organization throughout the estimate development stage.
- Have a plan for completion of the estimate
- Maintaining up-to-date files
- Estimate should be neat, clear, and easy to follow

Planning the estimate

- When will the work be done
- Who is responsible
- Diagrammatic representation of the steps that are required to complete an estimate (Fig. 4.1)
- Prepare Bar chart schedule for completing an estimate (Fig. 4.2)

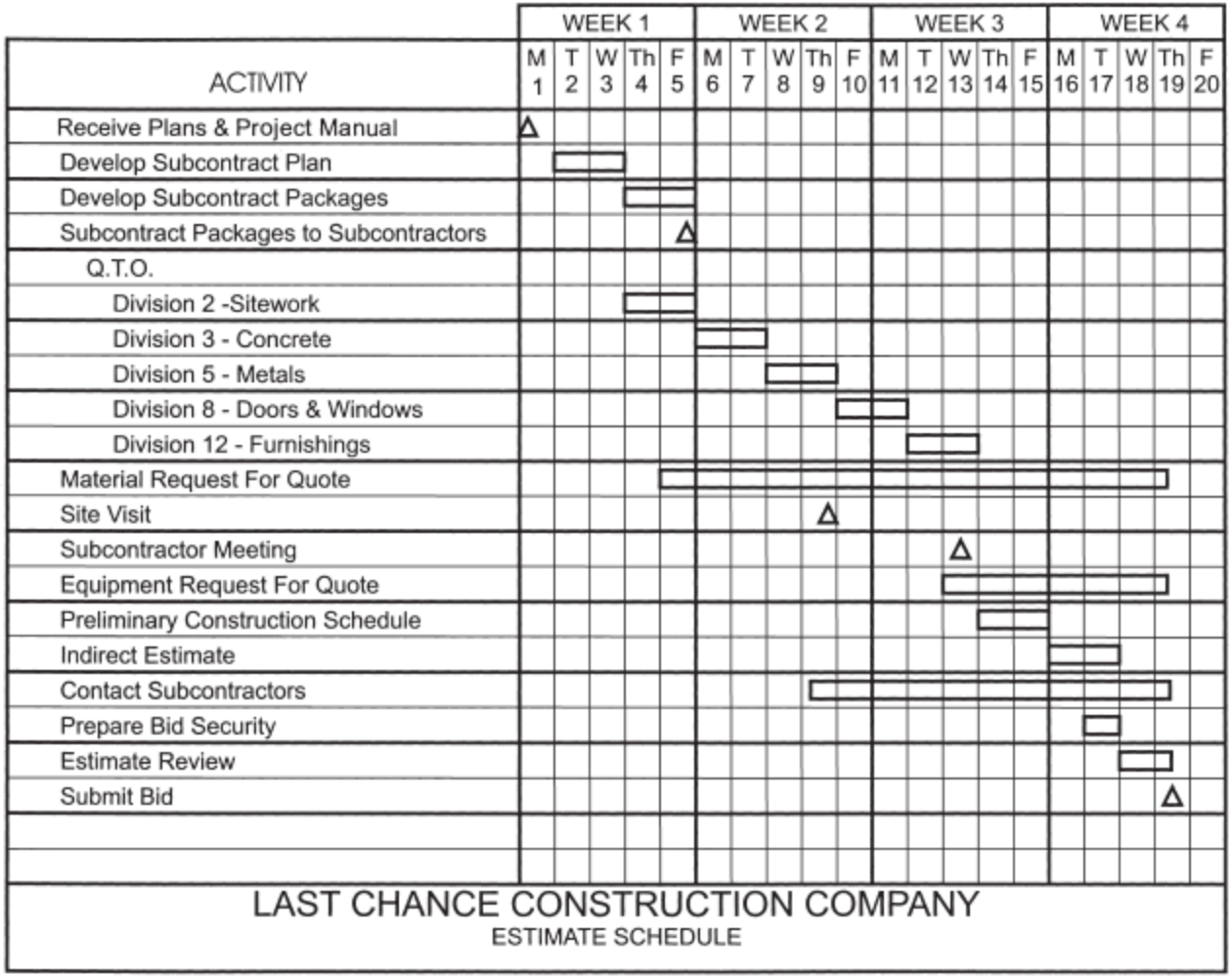
The Estimating Process

Figure 4.1



Bar Chart Schedule

Figure 4.2



Estimating Notebook

A notebook should be kept for each estimate prepared that may be broken down into:

- Workup sheets (every page must be numbered and initialed)
- Summary sheets
- Errors and omissions
- Proposals received from
  - Material suppliers and manufacturers
  - Subcontractors
- Notes pertaining to the project
- Calls made to the engineer

To Bid or Not To Bid

- Decision based upon:
  - Type of construction
  - Location
  - Size of project
  - Bonding capacity
  - Architect/engineer
  - Work in progress (in hand)
  - Availability of equipment
  - Availability of qualified personnel

The Estimate

If decision is made to bid then:

- Check the drawings and project manual for completeness
- Get a feel for the project
- Review the floor plans, room layouts
- Review wall sections to see what materials are being used
- Review structural drawings (prefixed with letter S)

The Estimate

- Review mechanical (plumbing and HVAC) drawings
- Read and study the project specification
- Visit the site
  - Take pictures
- Review general conditions and supplemental general conditions

The Estimate

- Order:
  - Insurance
  - Bonds
- Take off quantities using workup sheets
- Ask subcontractors and materials suppliers to bid
- List all overhead items required

The Estimate

- Summarize costs from workup sheet on the summary sheet
- Check for errors
- Verify with architect/engineer:
  - That you have all of the addenda
  - Time and place of bid

Common Error to Check For

- Math errors (+, -, x, /)
- Omission of items (material, labor, equip, overhead)
- Time to complete the project
- Errors in estimating construction waste
- Errors in estimating quantities of material
- Transferring number from one sheet to another

Common Error to Check For

- Adding a line to a spreadsheet and not including it in the sum
- Set up errors in software
  - Formulas
- Improper use of software

Site Investigation (Visit)

- Site access
- Utilities
- Drainage
- Transportation facilities
- Required protection or foundation of adjacent property
- Storage and equipment location
- Soil conditions
- Local ordinances
  - Permits
  - Licenses
  - Fences
- Local labor and union rules

Site Investigation (Visit)

- Local material and deliver pricing
- Subcontractor availability
- Road conditions to the project
- Housing and food facilities
- Banking facilities

Specialty Contractors

- Is a separate subcontractor hired by the prime contractor to perform certain portions of the work. E.g. plumbing, electrical, HVAC
- Advantages:
  - Less direct-hire craft personnel
  - Reduced risk
- Disadvantage:
  - Less control
- Bid Tabulation: Subcontractors bid + adjustments (see Fig. 4.4 for Subcontract)

Materials

- Quote should include:
  - Material cost
  - Freight (cargo, shipment)
  - Taxes
  - Delivery time
  - Terms of payment
  - Material Price Quote (Fig. 4.5)

Estimating Sheets

- Workup sheet
  - Used to make "work up" the cost of each item (Fig. 4.6 Workup Sheet – used to quantity reinforcing Steel
- Takeoff be complete, do not write e.g. "wire mesh" but "wire mesh 6x6
  - If mesh is galvanized, it will increase your material cost by about 20%

Estimating Sheets

- Summary sheet
  - Summarized cost on workup sheets
  - List all the information required
    - but none of the calculations and sketches of the workup sheet.
  - Figure 4.7 is an example of a summary sheet for concrete in the project.
  - Figure 4.8 summarize all cost for the project.

Errors and Omissions

- Make list of errors and omissions
- Get clarification from architect/engineer
- Specifications take precedence over drawings and dimensioned figures
- Detailed drawings take precedence over scaled measurements from drawings.

Thank You